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**B.I.D.A (Bridging Inclusion through Digital Accessibility): A
Web-Based Digital Portal for E-Services and Information Access
for Dasmariñas City Social Welfare and Development Office**

AN UNDERGRADUATE CAPSTONE PROJECT

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Bachelor of Science in Information Systems**

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CHAPTER 1 INTRODUCTION

Project Context

The existing public service infrastructure of the City of Dasmariñas still mostly relies on traditional or paper-based processes. This can limit the efficiency, transparency, and the overall accessibility of these essential services. These outdated processes constitute a significant problem, specifically to the elderly and PWDs. They often experience inconvenience with needing to travel from different parts of the city, hurdles with program participation, timely updates, and in requesting city services and transactions. According to Syahirah Zulkifli et al., (2023), improper planning of record management in an organization can lead to difficulties of long-term preservation and information retrieval in future use.

Additionally, the lack of a centralized digital portal makes it difficult for the said sector to access for updates, submit feedback, request assistance, and take advantage of support programs by the city government. This dilemma is leading to inequity, diminishes community participation, and strains the overall function of citizens and city government. An effective government website allows its residents to quickly perform tasks online, apply for permits, and request for documents which avoids long wait times compared to visiting in person (The Diligent Team, 2020).

In order to address these issues, we aim to create a web-based system designed to provide a unified platform for citizens of Dasmariñas under the CSWDO sector. The proposed system will grant secure, user-friendly digital access to essential city services and existing user



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data. Digitalizing services is expected to offer a faster, flexible, and convenient way for residents, including PWDs and senior citizens, to request assistance and participate in programs without facing physical barriers (Castillon, Magallanes & Presbitero, 2023). By integrating accessibility features such as voice navigation, text-to-speech, and intuitive navigation, we can foster digital inclusion, promote independence, and make it easier for everyone to access services and improve their quality of life. Overall, this proposal aims to promote responsiveness and inclusivity for the City of Dasmariñas.

Purpose and Description

The purpose of this study is to develop a web-based digital portal for PWDs and senior citizens in Dasmariñas. The goal is to streamline various processes and transition from a manual setup to a fast, secure, and digitized environment. This system aims to help CSWDO staff in accessing, storing, and retrieving documents uploaded by users. Additionally, it will improve service accessibility by allowing users to request services online, schedule appointments, and receive assistance in advance. By promoting the shift to a digital platform, the local government can reduce the physical barriers that are faced by PWD and Senior Citizens. This organized approach will also lessen the paperwork burden on the staff, allowing them to focus on improving the service quality given to their clients. Overall, our system seeks to promote a more inclusive and responsive public service system for the overall city welfare.



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Objectives of the Study

General Objectives

The primary objective of this study is to develop and implement B.I.D.A. (Bridging Inclusion through Digital Accessibility), a web-based digital portal for E-services and information access designed for the Dasmariñas City Social Welfare and Development Office to support the elderly and Persons with Disabilities (PWDs).

Specific Objectives

The specific objectives of this study will guide the development of the web-based digital portal. These steps follow a standard sequence to ensure the system is designed, built, and tested properly. The goal is to create a fully functional platform that will meet the needs of the target users and the CSWDO staff. Specifically, the researchers aim to do the following:

1. To design a web-based system that is capable of:

- Secure user registration and personal profile management.
- Digital service request submission and document routing.
- Accessibility features including voice navigation and text-to-speech.

2. To develop the system's frontend interfaces and backend database architecture utilizing web frameworks including HTML, JavaScript, PHP, and MySQL.

3. To test the portal's core functionalities, data security, and accessibility compliance to identify and resolve technical errors across various devices.

4. To evaluate the system's usability, performance, security, and efficiency through User Acceptance Testing (UAT) involving CSWDO personnel and target end-users (senior citizens and PWDs).



5. To implement the fully functional e-service portal on a secure live server, successfully integrating the automated SMS and call notification features into the existing government workflow.

Conceptual Framework

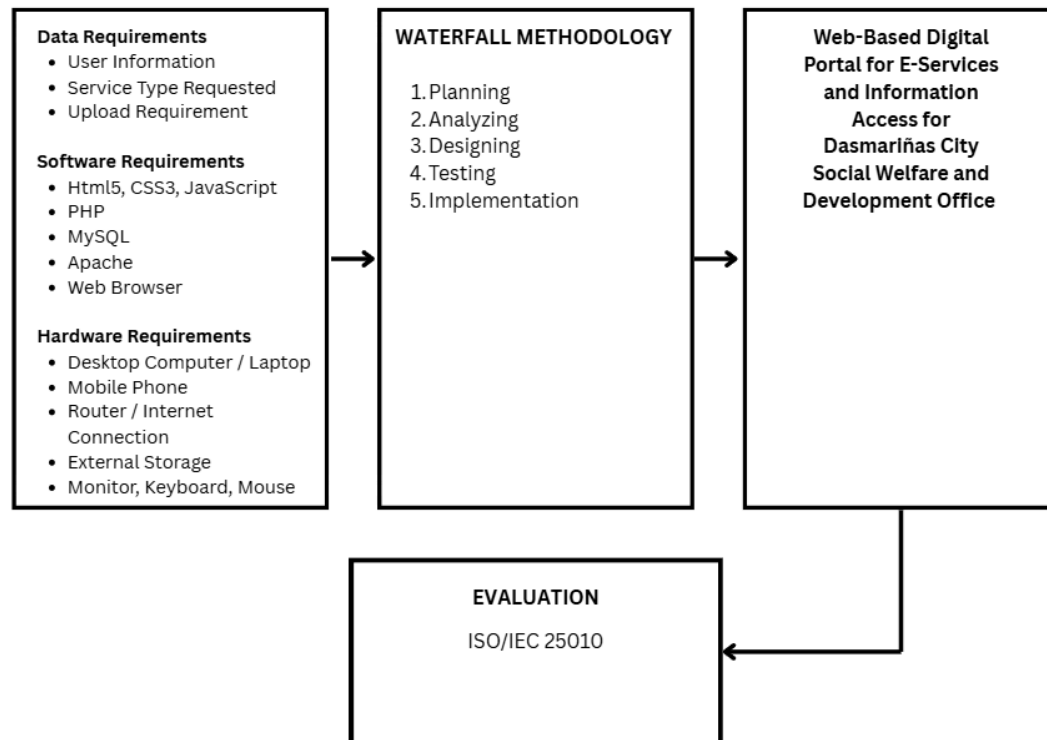


Figure 1

The conceptual framework of this study illustrates how the B.I.D.A. Portal functions as a system that transforms the delivery of services for Seniors and PWDs in Dasmariñas. The framework begins with data requirements which will include user information, the service type requested, and uploaded documents. We will also require the necessary hardware in building the system such as computers, mobile phones, internet connection, and other peripherals. Finally for our system requirements, we will build the system around HTML, CSS, and JavaScript for the front end, while MySQL and Apache will be utilized for our back-end.



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The process for building our system will involve several key phases starting with planning where the needs and goals are identified, analyzing where we will be focusing on studying the current workflow and gathering necessary requirements, design where UI wireframes, database schema, and system architecture are created, testing which includes functional and usability testing to ensure the system works as expected, and finally, implementation where our web based portal is deployed on a local for server side and online for client side.

As a result, the primary output of this study is the fully functional web-based platform designed to bridge the gap between digital convenience and physical service delivery. While the system does not entirely eliminate the manual release of tangible services such as IDs or financial aid, it ultimately creates pre-processed and verified transactions. By changing the time-consuming stages of filing, document submission, and status checking to a digital environment, the system ensures that the output of the online process is a 'ready-to-release' status. This design streamlines the workflow, which overall improves the level of accessibility for PWD and senior citizen sectors, greater transparency in tracking requests, and enhanced overall efficiency.

Scope and Limitations

The scope and limitations of this study define the operational boundaries of the B.I.D.A. Web-Portal. This section outlines the specific functionalities provided to the City of Dasmariñas and the technical or procedural constraints that fall outside the study's objectives.

- **Intended Users and Functions:**



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- **Primary Users:** Registered Senior Citizens and Persons with Disabilities (PWDs) residing in the City of Dasmariñas.
- **Administrative Users:** Authorized personnel of the City Social Welfare and Development Office (CSWDO) for data management and service processing.
- **Modules and Features:**
 - **User Management:** Secure registration, personal profile hosting, and digital document uploading.
 - **Service Processing:** Online submission for city services, automated document routing, and appointment scheduling.
 - **Notification System:** Integration of automated SMS and call alerts to provide real-time updates on application status and scheduled visits.
- **Research Focus:**
 - **Digital Accessibility:** Implementation of Voice Navigation and Text-to-Speech (TTS) features specifically designed for users with visual or motor impairments.
 - **Process Optimization:** Transitioning manual filing into a "ready-to-release" digital workflow to reduce physical wait times at the CSWDO.

Limitations of the Study

- **Platform Restriction:** The system is strictly a web-based application accessed via browsers; it does not include a native mobile application (APK/IPA).
- **Departmental Boundary:** The system is exclusive to CSWDO-managed services and does not integrate with other city departments (e.g., City Health Office or Assessor's Office).



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- **Exclusion of Other Sectors:** The portal does not cater to Solo Parents or youth scholars unless they are also registered as a PWD or Senior Citizen.
- **Financial Transactions:** The system excludes an online payment gateway. All financial aid disbursements or fee collections remain manual, in-person transactions.
- **Physical Verification:** While the system allows for document uploads, the final authentication of physical documents (e.g., original medical certificates) remains the responsibility of CSWDO staff.
- **Technical Dependencies:** The portal requires a stable internet connection and compatible hardware; it offers no offline functionality for end-users.



Definition of Terms

- **Android Package Kit (APK)** - The package file format used for distributing and installing applications on the Android Operating system.
- **CSWDO (City Social Welfare and Development Office)** - The specific government sector in the City of Dasmariñas responsible for managing social welfare programs and serving as the primary administrative user of the proposed digital portal.
- **Database Architecture** - The structural design of the system's data environment, utilizing MySQL to ensure that user information and document records are organized for secure and efficient retrieval by city personnel.
- **Database Schema** - The logical configuration or "blueprint" of the database that defines how data elements, such as user profiles and service request types, are categorized and related to one another within the system.
- **Digital Accessibility** - The practice of designing the web portal to ensure it can be used by everyone, specifically incorporating assistive features to remove barriers for the elderly and Persons with Disabilities (PWDs).
- **Digital Portal** - A centralized, secure web-based entry point that provides a unified platform for the citizens of Dasmariñas to access government information, submit documents, and request city services.
- **E-Services** - The electronic delivery of government functions, such as online registration, appointment scheduling, and automated status updates, intended to replace traditional, paper-based transactions.
- **Ready-to-Release Status** - An operational milestone within the system indicating that a user's application and uploaded documents



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have been digitally pre-verified, leaving only the final, brief in-person collection of physical aids or IDs.

- **Text-to-Speech** - An assistive technology feature of the B.I.D.A. portal that converts written on-screen text into audible speech, facilitating better navigation for users with visual or cognitive impairments.
- **UI Wireframes** - The visual blueprints or skeletal frameworks of the portal's user interface used during the design phase to establish the layout and navigation flow before the actual coding begins.
- **User Acceptance Testing (UAT)** - The formal evaluation phase where the system is tested by actual CSWDO personnel, senior citizens, and PWDs to ensure the portal meets their specific needs and operates efficiently.
- **Voice Navigation** - A hands-free accessibility feature that allows users to interact with the portal and navigate through different service modules using spoken commands.
- **Web-Based System** - A software application accessed through a web browser over an internet connection, eliminating the need for users to install local software on their devices.



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